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NEURAL BELLMAN OPERATORS

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Neural Bellman Operators construct and learn continuations for feasible decisions under a specified recursive economic objective. We connect their own-future policy account to action-dependent error, continuous-risk information and complete computational cost. For a convex Lipschitz continuation under uniform innovations, locating the adverse tails before integration gives matching deterministic point-information bounds of order $KR(1-\alpha)N^{-2}$, with risk-level-uniform constants. An outward original-model tail defect and global action supports yield a certificate for the implemented complete policy. A bounded prelude and packed evaluation make this information account operational before an unchanged complete fallback. Original Bellman foundations, constrained gradients, signed policy comparisons and recursive-utility boundary conditions receive full proofs. Separate frozen comparisons retain all neural, hinge, polynomial, spline, semi-analytic and grid controls, original-policy reference enclosures and actual failed or adverse evidence. Packing improves 10 of 16 contemporaneous median service-time cells against the geometric control, while every service returns the same original policy. The quadratic representation remains a complete-cost and economic winner in the reported catalogue. These results separate original economic gains from implementation savings and identify action-aligned learning and matched-cost adoption as the next substantive NBO questions.

KEYWORDS: Dynamic programming, Neural approximation, Recursive risk, Policy qualification, Computational economics.

1. INTRODUCTION

Dynamic investment under risk requires a continuation value, a feasible action, and a decision about how much information to acquire before acting. A small continuation fitting error need not imply a small decision loss. It can move a nearly

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binding action, alter the adverse innovation tail, or leave an expensive original-model verification problem unresolved. Neural Bellman Operators (NBOs) study these objects together: recursively constructed and learned continuation maps propose policies, their own future decisions determine policy value, and original-law Bellman inequalities qualify the actions actually delivered.

The economic question is when an acquired neural continuation helps deliver original-objective decisions at a competitive complete cost. Answering it requires conventional representations with the same information, feasible actions and numerical components. A conventional hinge encoding of the same ReLU map receives the same policy and certificate. A quadratic continuation that wins a complete-work comparison remains a successful control. The purpose of these comparisons is to identify which part of continuation construction, information acquisition and policy improvement creates economic value.

This paper develops a chain from recursive continuation construction to action-aligned information and verified decisions. The retained own-future NBO construction gives an exactly realizable neural continuation and a measurable feasible actor without assuming successful nonconvex training. Centered recursive-risk inequalities separate irrelevant value-level shifts from action-dependent approximation errors. Original-model supporting quadratics then connect the realized constrained action to the original Bellman optimum. The assumptions behind Bellman existence, measurability, continuity and uniqueness are stated explicitly; the supplement proves them for compact innovation supports and verifies the original investment primitives.

Nonlinear risk introduces an economic obstacle beyond directional fitting error. Even when two continuations have identical action derivatives at every innovation, their maximizing risk densities can differ. The retained continuous-law tail-switch example has zero directional approximation error and original action loss $1/4$. Centered quantile-band bounds identify the additional near-tail information needed to control this discrepancy. The present addition determines how much point information is needed to resolve that tail, and constructs a policy certificate when its location is only approximately known.

For a convex K-Lipschitz continuation on $[-R, R]$ under the original uniform law, an adverse tail of probability $t = 1 - \alpha$ consists of at most two endpoint intervals. Its complement is a central interval of fixed length. The derivative of the integral over that central interval is the difference between two endpoint continuation values; it is monotone and Lipschitz. Consequently paid bisection locates the relevant tails, and midpoint–secant integration can concentrate on their total length $2Rt$. Including every localization query, we prove a deterministic worst-case risk-width upper bound of order $KRtN^{-2}$. Together with a fully restated indistinguishability lower bound, this gives matching risk-level dependence for $N \geq 32$. This resolves the factor t^{-1} in the retained whole-support upper account. The theorem specifies exact point information; bit precision and the work needed to obtain each continuation value have their own implementation account.

The original nonlinear investment economy supplies an implementable version. Primitive curvature implies that the optimal continuation along its innovation direction is strongly convex with modulus $181/64$. For any proposed feasible tail cut, outward intervals for the two endpoint values bound its failure to maximize the original risk functional by a constrained quadratic calculation. This defect is paid explicitly. The same fixed feasible density and global continuation supports produce a lower bound over the entire original action interval. Selected-tail secants and the unchanged implemented-future allowance produce an upper bound on the actual complete policy. No derivative through a rounded actor, fitted sign assertion or hypothetical future optimizer obtains certificate authority.

The computational contribution is therefore a change in which original continuation values are purchased and how their errors enter a decision, rather than an internal fit-score improvement. A geometric precision schedule can stop at the first qualified resolution, while a complete fallback handles an intrinsically inadequate candidate or supporting minorant. The retained completed-incumbent rule gives strict original-policy improvement. Its remaining-gain budget bounds the economic improvement sacrificed by stopping further search. A coupled supplementary account integrates common-density policy comparisons, atom-safe tail remainders and signed recursive gain transport, with the independently frozen earlier research on that question credited in the development companion.

The numerical study holds the original two-date investment technology, continuous innovation distribution, nested AVaR preference, feasible actions, terminal controller and accuracy targets fixed. Eight arms share fresh paid terminal-value labels: the online semi-analytic Bellman control, a rank-six quadratic, ReLU, its equivalent conventional hinge, coordinate splines, full quadratic and cubic polynomials, and tensor-grid interpolation. Each arm receives the same candidate policies under each contemporaneous certificate service. Three separately frozen phases retain 32,256 primary decisions and 50,176 fresh-process timing decisions, together with acquisition, fitting audits, rejected attempts, repairs, original-law reference calculations and failures. Phase A establishes the active-tail implementation; its adverse elapsed-work comparison motivates phase B's bounded preliminary acquisition. Phase B's remaining overhead motivates phase C's packed original evaluation. The latter two phases are explicit development on the previously observed catalogue. Every control is executed afresh, and neither phase replaces the observations of its predecessor.

Phase C's packed service is faster than its uncombined budgeted control in all 16 median-clock cells and faster than the retained geometric service in ten, with all six losses reported. The ReLU service-time ratios against the geometric control are 0.8614 and 0.8494 at the two risk levels. Returned policies are unchanged across services, so their economic cost contrasts are exactly zero. With fresh standalone acquisition and actual model loading charged, the quadratic control still has the lower original mean economic cost and lower complete component cost at both risk levels. Conventional hinge and ReLU maps remain policy-identical. The fixed catalogue is a reproducible numerical comparison; its results carry no population-sampling interpretation. Seven fresh timing processes characterize observed machine variation and do not supply sampling confidence intervals.

The contribution builds on established risk-averse dynamic programming and policy iteration (Ruszczyński, 2010, 2014), the AVaR variational formulation (Rockafellar and Uryasev, 2002), and information-based integration (Novak, 2015). Inexact-oracle optimization and decision-focused learning clarify related accuracy and acquisition questions (Devolder et al., 2014, Elmachtoub and Grigas, 2022). Here the mathematical addition is the joint tail-location and integration account with

matching tail-length dependence, and its outward implementation as an all-action certificate under the same original economic law. The compiler and these certificates are available to every representation. Formula information and derivative access are charged and distinguished from exact point values.

The original NBO program also concerns stochastic control and nonlinear recursive utility. A Brownian Markov BSDE verification theorem states an admissible connection, including driver measurability, the critic graph $z = \sigma^{\mathsf{T}} \nabla u$, actor-chain derivatives, terminal data and stopping-boundary errors. General NDU/FBSDE systems require their own existence, information and objective assumptions. The investment evidence is attached to its nested AVaR objective, while the recursive-utility theorem is attached to the utility problem it actually proves.

The complete article retains earlier original-policy improvements and their economic stopping costs, the historical quadratic complete-work winner, all 36 improved residual spans accompanied by an increase from 644 to 757 first-proposal rejections, and the unestablished neural-adoption regions. These observations motivate action-aware acquisition and stronger numerical methods. Current progress is evaluated by new proofs, complete decisions and paid work. Neural-specific representation advantages, prospective acquisition validation, longer horizons and calibrated economic adoption remain distinct research obligations. The supplement supplies complete mathematical arguments; the development companion and source archive preserve earlier theories, applications, adverse evidence and independently frozen experiments.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t, the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$ and $B_t = \prod_{j<t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v, define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \qquad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \tag{1}$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the earlier expected-cost experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. Section 5 evaluates the same investment technology under an explicitly nested AVaR preference and retains the original continuous innovation law. The direct-cost inference in Section BD is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It forms targets using its *own* fitted f_{t+1}, returns a neural continuation and a feasible policy, and records an economic error allowance. It does not receive the optimal continuation as a training label. Arbitrary trained networks can be evaluated by the first result below. The subsequent constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. *Action-dependent errors and complete-policy loss*

Let $H(a)$ be a true current cost and $\widehat{H}(a)$ its continuation approximation. If $\widehat{a}$ is δ-optimal for $\widehat{H}$ on the same feasible action set, then

$$H(\widehat{a}) - \inf_a H(a) \leq \delta + \operatorname{osc}_a\{\widehat{H}(a) - H(a)\}. \tag{2}$$

Indeed, compare $\widehat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No

transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \le u_t, \qquad \mathcal{T}_t^\pi f_{t+1} - f_t \le v_t, \qquad \ell_T \le f_T - g \le u_T.$$

Then the actual feasible policy satisfies

$$0 \le J_0^\pi - V_0^* \le \sum_{t<T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \tag{3}$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

The proof is in Supplement, Section 1.1.

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state–action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \le K_t^A \|x - z\|_1, \qquad L_t = A_t^Q + D_t K_t^A, \qquad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t-Lipschitz whenever f_{t+1} is L_{t+1}-Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t, form effective feasible k_t-nets. Evaluate their own-future objectives with certified error at most e_t. Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \le K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \tag{4}$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t\|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg\min_i \{y_{t,i} + L_t\|x - x_{t,i}\|_1\}. \quad (5)$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g.

THEOREM 2—Own-future construction with feasible witnesses: *The functions in* (5) *have exact affine–ReLU realizations and Lipschitz moduli L_t independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \le f_t - \mathcal{T}_t f_{t+1} \le D_t k_t + e_t + 2L_t h_t, \quad (6)$$
$$\mathcal{T}_t^{\pi} f_{t+1} - f_t \le e_t + D_t \zeta_t,$$

and hence

$$J_0^{\pi} - V_0^* \le \sum_{t<T} B_t(D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T(2e_T + 2L_g h_T). \quad (7)$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

The proof is in Supplement, Section 1.2.

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are *one method with two encodings*. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

4. REFERENCE-CENTERED RISK KERNELS AND COMPLETE-POLICY QUALIFICATION

A datewise worst-case error account can charge incompatible states at successive dates. Under nonlinear recursive preferences, replacing that account by an

expectation under the physical transition law is not valid. We instead retain the conditional risk evaluation and center its dual representation at the actual stored NBO continuation. The resulting state-dependent certificate has an original-optimum interpretation and a finite occupation representation. It extends, rather than replaces, the signed qualification of Section Z.

4.1. *The centered operator*

Retain the original augmented economic state, feasible actions, nonnegative discounts, and bounded complete-policy recursions. At each (t, x, a) suppose the original conditional evaluation admits the representation

$$\rho_{t,x,a}(v) = \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int v(y) q(dy) - \psi_{t,x,a}(q) \right\}. \tag{8}$$

Every q is a probability measure on the original successor state space. The representation is assumed on a common bounded-function domain containing all continuations below, with the measurability and selection conditions needed for the original Bellman recursions. This is a hypothesis about the original preference primitive, not permission to replace a static risk objective by a nested one. Conditional risk duality and risk-averse dynamic programming are established principles (Ruszczyński, 2010, 2014). For a nonconvex or non-cash-invariant aggregator lacking (8), the earlier order and modulus results remain applicable.

Fix one qualified parameter θ and suppress its subscript. Its entire sequence of stored continuations is $f = (f_t)_{t=0}^{T}$, and its supplied complete feasible policy is π. Define, using the same f_{t+1} in every term,

$$\begin{aligned} r_t(x, a) &= c_t(x, a) + \beta_t \rho_{t,x,a}(f_{t+1}) - f_t(x), \\ k_t^f(x, a, q) &= \psi_{t,x,a}(q) + \rho_{t,x,a}(f_{t+1}) - \int f_{t+1}\, dq, \\ \mathcal{E}_{t,x,a}^f(z) &= \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int z\, dq - k_t^f(x, a, q) \right\}. \end{aligned} \tag{9}$$

The penalty is nonnegative and has infimum zero, whether or not the supremum in (8) is attained. Direct substitution gives

$$\mathcal{E}^f_{t,x,a}(z) = \rho_{t,x,a}(f_{t+1} + z) - \rho_{t,x,a}(f_{t+1}). \tag{10}$$

In particular, $\mathcal{E}^f(0) = 0$, and $\mathcal{E}^f$ is monotone, cash invariant, and nonexpansive in the uniform norm. Discarding k^f is an upper relaxation, not this identity. Freezing one risk-maximizing kernel at f is in general not even an upper relaxation.

THEOREM 3—State-dependent original-optimum certificate: *Suppose verified functions satisfy $r_t^-(x,a) \le r_t(x,a)$ for every original feasible action and $r_t(x,\pi_t(x)) \le r_t^{\pi,+}(x)$. Let bounded measurable L_t, U_t satisfy*

$$\begin{aligned} L_T &\le g - f_T \le U_T, \\ L_t(x) &\le \inf_{a \in A_t(x)} \{r_t^-(x,a) + \beta_t \mathcal{E}^f_{t,x,a}(L_{t+1})\}, \\ U_t(x) &\ge r_t^{\pi,+}(x) + \beta_t \mathcal{E}^f_{t,x,\pi_t(x)}(U_{t+1}). \end{aligned} \tag{11}$$

Then, at every initial state,

$$f_0 + L_0 \le V_0^* \le V_0^\pi \le f_0 + U_0, \qquad 0 \le V_0^\pi - V_0^* \le U_0 - L_0. \tag{12}$$

The same statements hold at every intervening date and state. They are uniform over a qualified parameter class whenever all premises and the reported bound hold for each member with its own common continuation sequence and complete policy. The certificate permits endogenous states and actions changing at every date; it does not require a finite state space or independent deployment contexts.

With exact residuals and terminal discrepancies, equality recursions in (11) give $L_t = V_t^* - f_t$ and $U_t = V_t^\pi - f_t$. With scalar bounds from (A.28), their corresponding constant lower and upper sequences are feasible certificates here. Thus the exact state-dependent recursion is no looser than the signed scalar account. A strictly smaller bound can result without changing a fit, a feasible action, or an economic observation.

The last assertion concerns the best exact recursion or a verified improvement upon the constant certificates. It is not a claim that every approximately computed

L, U improves the scalar account. In implementation, the minimum of valid policy supersolutions and the maximum of valid optimum subsolutions retain validity, so an inferior new bound need not displace the existing one.

The lower inequality requires every original action. A discrete action search without a verified original-menu lower bound cannot establish it. For a fixed action, a feasible risk kernel supplies a lower bound on $\mathcal{E}^f(L)$; an upper bound requires control of the entire risk envelope. Numerical integration, feasible repairs, residual errors and penalty errors belong in these inequalities. If penalty enclosures are used, lower penalty endpoints enter an upper certificate and upper endpoints enter a lower witness. Approximate dual optimizers are accepted only through these independently checked inequalities.

5. CONTINUOUS-LAW QUALIFICATION IN THE INVESTMENT MODEL

The risk theorem becomes a numerical method only when its inequalities can be constructed from economic primitives. We now supply this construction for the original investment technology with a nested tail-risk preference. States and innovations remain continuous, every feasible action is covered, and the implemented terminal controller rather than a hypothetical exact optimizer determines policy cost. This application joins continuation construction, the original-optimum comparison and a representation comparison on one problem. It does not replace the general NBO construction or the other economic applications.

5.1. *Primitives and the object being approximated*

Let $d \geq 2$ be even, $X = [0,1]^d$, and interpret subscripts cyclically. At dates $t = 0, 1$, the feasible investment is $0 \leq a \leq A(x) = 1/8 + \bar{x}/8$, where $\bar{x} = d^{-1} \sum_j x_j$. Write $b_j = 1/2$ for odd j and $b_j = 1/4$ for even j, and $s_j = (-1)^{j+1}$. The original transition is

$$F_j(x,a,z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1-x_{j+1})}{16} + b_j a + s_j z, \qquad z \sim \mathrm{Unif}[-1/32, 1/32]. \tag{13}$$

Define

$$S(x) = \frac{2}{d}\sum_j (x_j - 5/8)^2 + \frac{1}{4d}\sum_j (x_j - x_{j+1})^2 + 2(1/2 - 2\bar{x})_+^2, \tag{14}$$

$$g(x) = S(x) + \frac{2}{d}\sum_j (x_j - 5/8)^2, \qquad h(a) = a^2 + 4a^4.$$

We use $\beta = 15/16$ and, separately, $\alpha \in \{1/2, 3/4\}$. Average value at risk is the upper-tail cost functional

$$\rho_\alpha(v) = \sup\left\{\int vq\,dP : 0 \le q \le (1-\alpha)^{-1},\ \int q\,dP = 1\right\}.$$

Here P is the original uniform innovation law. The objective is recursively defined by $V_2 = g$ and

$$V_t(x) = S(x) + \inf_{0 \le a \le A(x)} \{h(a) + \beta\rho_\alpha(V_{t+1}(F(x,a,\cdot)))\}.$$

It is not the static tail risk of total realized cost. A complete policy value replaces each infimum by the action prescribed by that policy, with the same recursive risk primitive.

Assumptions for the continuous construction. For this subsection, the primitives are exactly (13)–(14); α is fixed before construction; the fitted continuation f is a finite convex function on X; and every numerical lower or upper value is an outward enclosure. Feasible tail densities and feasible actions are retained explicitly. Terminal selection is the measurable, dyadically acquired controller described below, with its fallback included in the error bound. These concrete conditions imply compact feasible graphs, bounded costs, measurable minimizers, and the conditional-risk dual used in Theorem 3. They do not assert that every recursive preference or every trained network has this convex structure.

PROPOSITION 4—Primitive curvature and state supports: *For either risk level and every even d, the one-date objective $Q_1(x,a) = S(x) + h(a) + \beta\rho_\alpha(g(F(x,a,\cdot)))$ is jointly convex on $X \times [0, 1/4]$, with curvature bounded below by*

$$\frac{4 - 5\beta/4}{d}\|v\|_2^2 + 2w^2 \tag{15}$$

in a direction (v, w). Its value V_1 is convex on X and satisfies

$$|V_1(x) - V_1(y)| \le \frac{15}{d}\|x - y\|_1. \tag{16}$$

Let W be the objective evaluated under any fixed feasible risk density, at an anchor (x_0, a_0), and let (p_x, p_a) be its gradient there. For any $\lambda, \nu \ge 0$, set $e = p_a + \lambda - \nu$. Then the following is a global affine minorant of V_1:

$$\ell(y) = W + \lambda(a_0 - A(x_0)) - \nu a_0 - \frac{e^2}{4} + \left(p_x - \frac{\lambda}{8d}\mathbf{1}\right)^\top (y - x_0). \tag{17}$$

Neither optimal multipliers nor an exact optimizing action are required.

The curvature is obtained from the original nonlinear technology, not imposed on an interpolated value grid. The negative second-order term from $x_j(1 - x_{j+1})$ is bounded using $|\partial_j g| \le 10/d$; the stage curvature dominates it. The allowance $e^2/4$ pays for imperfect stationarity. The proof in the Technical Supplement also establishes the global state modulus in (16), including the state-dependent action constraint.

5.2. *An all-action certificate under the original innovation law*

Partition the innovation interval into n equal bins B_i, with midpoints z_i and probabilities $p_i = 1/n$. An innovation bin is a set over which an inequality holds, not a replacement point mass in the economic model. Choose nonnegative $q_i \le p_i/(1 - \alpha)$ summing to one. These masses define a feasible density constant on each bin. For a fixed initial state and feasible proposal a_0, let W_0 be a lower bound on the objective under this density, obtained from globally valid affine supports of the continuation at the successor midpoints. Let p enclose the derivative of the same supporting objective at a_0. Then

$$\underline{V}_0(x) = \inf_{0 \le u \le A(x)} \{W_0 + p(u - a_0) + (u - a_0)^2\} \tag{18}$$

is a lower bound on the original Bellman optimum when the supported continuation is V_1. The scalar quadratic is minimized over the full original interval. Interval uncertainty in p is covered by taking the smaller minimum at its two endpoints, since the minimum is concave in p.

For a convex fitted continuation, midpoint supports give the lower risk witness and the larger endpoint value bounds the continuation throughout each bin. For V_1, Proposition 4 supplies midpoint supports and the allowance $15/(32n)$ covers each half-bin. Upper tail risk is the average of the largest $n(1-\alpha)$ upper bin bounds. This average is the exact risk of a dominating step function, hence an upper bound under the continuous law.

THEOREM 5—Constructed original-optimum qualification: *Suppose the supports and bin bounds above are outwardly verified. Suppose further that the complete terminal controller satisfies $V_1 \le V_1^{\widehat{\pi}} \le V_1 + \varepsilon_1$ everywhere on X. For every queried initial state, the lower value (18) and the binwise upper bound $\overline{J}_0^{\widehat{\pi}}$ satisfy*

$$0 \le J_0^{\widehat{\pi}}(x) - V_0(x) \le \overline{J}_0^{\widehat{\pi}}(x) - \underline{V}_0(x). \tag{19}$$

For any stored convex continuation f, the same lower and upper constructions are admissible bounds for the centered-risk comparison of Theorem 3. Their validity does not depend on the representation of f, on a successful fit, or on the proposal being an exact minimizer.

To see the connection, write a verified continuation bracket as $\ell \le V_1 \le u$ and use errors $L = \ell - f$, $U = u - f$. The centered identity gives

$$\rho_\alpha(f) + \mathcal{E}^f(L) = \rho_\alpha(\ell), \qquad \rho_\alpha(f) + \mathcal{E}^f(U) = \rho_\alpha(u).$$

Joint evaluation avoids subtracting and then adding two separately rounded copies of $\rho_\alpha(f)$. The bound (18) uses one feasible risk density to minorize the required all-action infimum; the policy upper bound controls the full risk envelope. Thus the application constructs sufficient inequalities for the original theorem. It does not claim that a non-neural continuation would receive a different validity theorem, or that computing the original-value witness is free.

The terminal controller observes each state coordinate rounded down to 40 binary places. It accepts its floating proposal only after an outward all-action gap of at most 2^{-21} at that observed state; otherwise it uses rational bisection. The original-state transfer is less than $30\,2^{-40}$. Consequently $\varepsilon_1 = 2^{-20}$ covers the complete controller, including observation and fallback. Policy evaluation encloses this controller at every possible successor, not only at the midpoint states used by the integrator.

Well-posed original decisions. The comparison uses a specified conditional nested recursion on the original observation filtration. A sufficient primitive model has a compact state space, nonempty compact feasible sets varying continuously with state, continuous costs and invariant transitions, compact innovations with their original fixed laws, and finite monotone cash-invariant risk evaluations. These assumptions make the Bellman operator preserve continuous functions. For implemented Borel policies, the risk evaluation must also preserve Borel parameter kernels; the supplement proves this property for expectation and AVaR. Proposition 2 proves, rather than assumes, the resulting finite-horizon value existence and uniqueness, attained Borel minimizers, comparison against all feasible adapted history policies, and the dated stability bounds. These are classical foundations (Ruszczyński, 2010, 2014), not a claim of a new fixed-point principle. Pointwise transition continuity with an unbounded innovation support is insufficient for arbitrary monetary risk; the exact essential-uniform alternative is stated in the proof.

The original investment primitives satisfy these conditions directly: $F(x,a,z) \in [1/32, 27/32]^d$, $1/8 \le A(x) \le 1/4$, and all costs are bounded. Their joint curvature gives V_1 state strong-convexity modulus $(4 - 5\beta/4)/d$ and root action modulus two. Both dates therefore have unique continuous constrained optimal actors. Proposition 3 also derives the continuous value gradient with its capacity multiplier, the original-risk root derivative, and the actual observation/feasibility allowance. Differentiating a rounded actor is not used to establish any of these claims. The supplement records the full smooth critic–actor graph and distinguishes its gradient from a frozen-target update.

The separate Brownian recursive-utility bridge has its own existence and verification assumptions. Proposition 4 supplies the stopped-domain version, charging

the original stopping boundary and terminal condition. Its Lipschitz driver can depend on the backward value, in which case cash invariance is not automatic. The continuous-time stability factor and smooth critic graph are therefore verified within that original recursive criterion.

6. ACTION-ALIGNED ACQUISITION AND COMPLETE-POLICY WORK

A continuation is acquired to make a decision, not to minimize an approximation statistic in isolation. The retained investment experiment makes the distinction operational: every certificate-oriented fit reduces its validated residual span, whereas aggregate first-proposal rejections increase. We now construct an acquisition target in the units of the admission inequality. The construction uses the same original action set and risk law as the NBO policy account.

6.1. *A computable directional certificate*

Fix a state and date, and suppress their indices. Let $A \subset \mathbb{R}^k$ be nonempty, compact and convex, let Q be the original action objective, and let $a \in A$ be the actually realized proposal. A verified supporting calculation supplies $L, U \in \mathbb{R}$, $s \in \mathbb{R}^k$ and $\mu > 0$ such that

$$Q(a) \le U, \qquad Q(b) \ge L + s^{\mathsf{T}}(b-a) + \frac{\mu}{2}\|b-a\|_2^2 \quad \text{for every } b \in A. \tag{20}$$

These inequalities need not identify s as a derivative of Q. In particular, an imperfect risk witness or continuation minorant can lower L while retaining a useful supporting slope. Put $w = U - L$ and

$$\Gamma(a) = w - \min_{b \in A} \left\{ s^{\mathsf{T}}(b-a) + \frac{\mu}{2}\|b-a\|_2^2 \right\}. \tag{21}$$

A certified lower bound on the displayed minimum gives a valid, possibly larger, Γ. An unverified optimizer status does not.

THEOREM 6—Directional certificate for an NBO proposal: *Under* (20),

$$0 \le Q(a) - \inf_{b \in A} Q(b) \le \Gamma(a).$$

Let $\widehat{s}$ be a supporting slope of the surrogate used to propose a, let $N_A(a) = \{n : n^\mathsf{T}(b-a) \le 0 \ (b \in A)\}$, and suppose $\|s - \widehat{s}\|_2 \le d$. If a supplied $n \in N_A(a)$ satisfies $\|\widehat{s} + n\|_2 \le r$, then

$$\Gamma(a) \le w + \frac{(d+r)^2}{2\mu}. \tag{22}$$

For $A = [0, A_0]$, the exact minimum in (21) is attained at $b = \mathrm{proj}_{[0,A_0]}(a - s/\mu)$.

The proof, including boundary actions, is in Supplement 5. Equation (22) separates three trainable or computable quantities: the original-objective enclosure width w, the directional mismatch d, and the surrogate stationarity residual r. The residual span of the continuation is not one of these quantities. A readout can improve that span and worsen the supporting slope at a nearly binding action. Conversely, constant changes in a continuation need not affect any of them after consistent centering.

The supporting calculation is available under the original recursive-risk primitives. Suppose $F(b,\omega) = F(a,\omega) + G(\omega)(b-a)$, the stage cost is μ-strongly convex, and the risk dual has an action-independent feasible witness q on the original innovation space. If

$$V(F(b,\omega)) \ge \ell(\omega) + v(\omega)^\mathsf{T} G(\omega)(b-a),$$

then (20) holds with

$$L = c(a) + \beta\left\{\int \ell\, dq - \psi(q)\right\}, \qquad s = \nabla c(a) + \beta \int G^\mathsf{T} v\, dq, \tag{23}$$

and any valid upper bound U for the original risk objective. Integration and penalty errors must be enclosed in these quantities. The witness must remain feasible for every action being minorized; a witness feasible only at a is insufficient when the risk set itself depends on the action.

In the investment economy, $G = (1/2, 1/4)^\mathsf{T}$ at $d = 2$, the innovation law and AVaR cap are action independent, and $h'' \ge 2$. Thus $\mu = 2$ gives exactly the all-action quadratic of (18). The added result is the explicit link from an acquired directional support to its admission gap, rather than a new economic model or an assumption that an acquired value support is an exact derivative.

7. AN ACTIVE-REGION COMPILER FOR THE ORIGINAL BELLMAN ORACLE

A learned continuation must compete with the structure available in the original economy. In (13)–(14), that structure includes an algebraic last-date Bellman oracle. Giving this oracle only to a neural method would confound representation with information. We derive it explicitly and give its implementation to every comparator.

Put $R = 1/32$, $t = 1 - \alpha$, $r_j = F_j(x, 0, 0)$, and

$$H_x = \tfrac{1}{2} - 2\bar{r}, \qquad k_x = \frac{10}{d}\sum_j s_j r_j,$$

$$C_x = \frac{8}{d}\sum_j b_j(r_j - 5/8) + \frac{1}{2d}\sum_j (b_j - b_{j+1})(r_j - r_{j+1}).$$

The following formula uses the original continuous law, not a fitted transition or a finite-shock replacement.

THEOREM 7—Exact tail algebra and constrained Bellman regions: *For* $z \sim \mathrm{Unif}[-R, R]$ *and* $0 < \alpha < 1$, *define*

$$\mathcal{R}_\alpha(k) = \mathrm{AVaR}_\alpha(5z^2 + kz) = \begin{cases} \dfrac{5R^2(1+\alpha+\alpha^2)}{3} + \dfrac{\alpha k^2}{20t}, & |k| \le 10Rt, \\ \dfrac{5R^2(4\alpha^2 - 2\alpha + 1)}{3} + \alpha R|k|, & |k| \ge 10Rt. \end{cases} \tag{24}$$

The two expressions and their first derivatives agree at their junctions. For every even d *and every original state* x, *the last-date action derivative is*

$$\partial_a Q_1(x, a) = 16a^3 + 2a + \beta\{C_x + 41a/32 - 3(H_x - 3a/4)_+\} + \frac{5\beta}{4}\,\mathrm{clip}\left(\frac{\alpha(k_x + 5a/4)}{10t}, -\alpha R, \alpha R\right). \tag{25}$$

It is continuous and strictly increasing, with slope at least two wherever its second derivative is defined. The interval $[0, A(x)]$ *has at most four regions separated by* $H_x/(3/4)$ *and the two solutions of* $k_x + 5a/4 = \pm 10Rt$. *On each region* (25) *is* $16a^3 + pa + q$, *with* $p \ge 2$. *There is a unique constrained minimizer: zero or capacity when the corresponding endpoint derivative has the appropriate sign, and otherwise the unique interior root. With*

$u = \sqrt{p/48}$, that root on its region is

$$a = -2u \sinh\left\{ \tfrac{1}{3} \operatorname{arsinh}(q/(32u^3)) \right\}.$$

Coefficient preparation costs $O(d)$ arithmetic; region identification and the number of scalar inversions are bounded independently of d.

The complexity statement counts real arithmetic and scalar inversions, not unit-cost exact transcendental evaluation. The implementation uses floating inversion only to propose a dyadic action. It then applies the unchanged outward original-law supporting certificate, including the state-capacity multiplier and its stationarity allowance. A failed compiled certificate pays for its failed attempt and the original terminal oracle. Consequently compiler error can change work but cannot acquire certificate authority. The full terminal policy used in nested policy-cost evaluation is not replaced by the compiler. In particular, more accurate optimal-continuation information and a different deployed future policy are distinct interventions.

Gradients and the computational graph. The last-date value is continuously differentiable on the relative interior of the state cube, with the one-sided boundary interpretation, and

$$\nabla V_1(x) = \nabla_x Q_1(x, a^*(x)) - \lambda(x) \nabla A(x), \qquad \lambda(x) = \mathbf{1}_{\{a^* = A(x)\}} [-\partial_a Q_1(x, a^*)]_+. \tag{26}$$

Within a smooth interior action region, $\nabla a^* = -\partial_{ax} Q_1 / \partial_{aa} Q_1$; within a strictly active endpoint region it is respectively zero or ∇A. No global differentiability of the actor across region changes is asserted. In particular, differentiating the rounded actor or treating a feasible witness slope as this exact gradient is incorrect. Supplement Section 8 proves the formulas, uniqueness, boundary rules and certificate invariance. The original general own-future construction remains Theorem 2; the new experiment isolates its acquired last-continuation slice, not a newly trained long-horizon network.

8. TAIL-AWARE ACTION ACCURACY

A critic can match action derivatives pointwise in the innovation and still propose the wrong risk-averse action. The missing quantity is the change in the adverse tail. This section refines the directional account in Theorem 6; it does not replace an original-law witness by a training statistic.

Fix a state and a realized action a. On the same original probability space let Z_a be the true continuation outcome and H_a an acquired surrogate. In a neighborhood of a, suppose both are almost surely differentiable with integrably dominated derivatives. At a assume their α-quantiles are unique, carry no mass, and cut off upper tails of probability $t = 1 - \alpha$, where $0 < \alpha < 1$. These are local assumptions, not automatic properties of every fitted ReLU. Choose a constant c and certified numbers $\epsilon, \delta \geq 0$ such that

$$\|Z_a - H_a - c\|_\infty \leq \epsilon, \qquad \|\nabla_a Z_a - \nabla_a H_a\|_\infty \leq \delta.$$

The constant is an anchor-level centering; no derivative of a data-dependent centering rule is taken. Write η for the true quantile, $B_\epsilon = \{|Z_a - \eta| \leq 2\epsilon\}$, and, for an arbitrary vector v_0,

$$G_\epsilon = \operatorname*{ess\,sup}_{B_\epsilon} \|\nabla_a H_a - v_0\|_2, \qquad m_\epsilon = \min\{P(B_\epsilon), 2\min(\alpha, t)\}.$$

Set $G_\epsilon = 0$ when $P(B_\epsilon) = 0$.

THEOREM 8—Centered tail-switch bound: *Under these hypotheses, whenever* $G_\epsilon < \infty$,

$$\|\nabla_a \operatorname{AVaR}_\alpha(Z_a) - \nabla_a \operatorname{AVaR}_\alpha(H_a)\|_2 \leq \delta + G_\epsilon m_\epsilon / t. \tag{27}$$

If $P(|Z_a - \eta| \leq r) \leq Cr^\kappa$ *for the relevant* r, *the tail term is at most* $G_\epsilon C(2\epsilon)^\kappa / t$. *If additionally* $\|\nabla_a H_a - v_0\|_2 \leq L|Z_a - \eta|$ *on* B_ϵ, *it is at most* $LC(2\epsilon)^{\kappa+1}/t$.

Suppose the original current objective Q *is differentiable and* μ*-strongly convex on a compact convex action set* A, *and its stage cost is unchanged in the surrogate* $\widehat{Q}$. *If* $n \in N_A(a)$ *and* $\|\nabla \widehat{Q}(a) + n\|_2 \leq r_a$, *then*

$$0 \leq Q(a) - \min_A Q \leq \frac{\{r_a + \beta(\delta + G_\epsilon m_\epsilon / t)\}^2}{2\mu}. \tag{28}$$

The proof localizes changed tail membership to a quantile band and uses the zero total mass of the difference between two maximizing densities. Thus a constant component of the surrogate action gradient cancels: tail *gradient oscillation*, rather than its overall magnitude, controls the switching cost. Under a positive margin exponent, joint control of the centered value error and directional error gives a rate for action loss. A finite validation sample does not itself verify C, κ or the essential suprema. At atoms or nonunique tails the stated differentiability theorem is unavailable; the feasible-density minorants of Theorem 6 remain the appropriate certification device. When Z_a uses an optimal continuation, (28) bounds the current-action gap; an imperfect implemented future policy requires its separate recursive allowance before interpreting the bound as complete-policy regret. Numerical witness slack is paid there through $U - L$, not silently dropped from (28).

An exact diagnostic. Let z be uniform on $[-1,1]$, $\alpha = 1/2$, $a \in [-1,1]$, and stage cost a^2. Take $Z_a = (a+1)z$ and $H_a = (a-1)z$. Their action derivatives are identically z, yet the original and surrogate optimal actions are $-1/4$ and $1/4$. Using the surrogate action incurs original loss $1/4$. This continuous-law diagnostic, independently integrated in rational arithmetic, rules out any general action-loss bound that vanishes with directional error alone. It is not the investment economy, a new population sample, or an empirical neural advantage. Its conceptual provenance in the earlier tail-aware research protocols is retained.

Acquisition target and novelty. For a family satisfying verified margin assumptions, a useful acquisition target is the bound in (28), with envelope slack and actual qualification work added. The earlier prospective finite-menu selection theorem then applies to complete services with that verified certificate. Our executed comparison below does not retroactively become a training experiment for this new loss: its frozen acquisition remains least squares. CVaR gradient estimation and decision-focused learning already have established formulations (Tamar et al., 2015, Luo and Delage, 2026, Elmachtoub and Grigas, 2022). The present result is a centered, tail-localized perturbation bound tied to the same constrained Bellman action and a valid complete-policy account. The static trajectory-CVaR objective of Luo

and Delage (2026) is not the nested AVaR preference used here; its empirical gains cannot be transferred between those objectives.

9. RESOLVING THE ACTIVE INNOVATION TAILS

The original continuous innovation law supplies information that an NBO must share with its strongest conventional competitors. Convexity of a continuation along that innovation implies that the adverse outcomes occupy at most two endpoint intervals. Locating these intervals before quadrature changes which continuation values must be acquired. In particular, the upper information bound need not pay a factor $(1-\alpha)^{-1}$ for resolving the entire innovation support when only a fraction $1-\alpha$ is adverse. The result below is a deterministic point-information theorem and an outward Bellman implementation. It does not assert a neural representation separation, a randomized optimum, or multivariate tractability.

9.1. *A tail-uniform point-information account*

Let $f:[-R,R]\to\mathbb{R}$ be convex and K-Lipschitz, where $R,K>0$; the information consists of exact point values, the original uniform law, and these class restrictions. The formula of f, its derivatives, and its integrals are not supplied. Write $t=1-\alpha$ and $L=2R\alpha$ for $0<\alpha<1$, and let $I=[-R,R-L]$. For $u\in I$ put

$$H(u)=\int_u^{u+L} f(z)\,dz, \qquad D(u)=f(u+L)-f(u), \tag{29}$$

$$R_u(f)=\frac{\int_{-R}^{u} f(z)\,dz+\int_{u+L}^{R} f(z)\,dz}{2Rt}. \tag{30}$$

Every R_u is a feasible original-law AVaR witness: its density relative to the original uniform law is t^{-1} on the two displayed intervals and zero elsewhere. Its feasibility does not require u to be optimal.

THEOREM 9—Active-tail information with uniform risk-level constants: *The function H is convex and continuously differentiable, $H'=D$, and D is nondecreasing and $2K$-Lipschitz. Moreover,*

$$\mathrm{AVaR}_\alpha(f)=\max_{u\in I} R_u(f). \tag{31}$$

If $D(-R) \geq 0$, the left endpoint is a maximizer in (31); if $D(R-L) \leq 0$, the right endpoint is one. Otherwise a zero of D lies in the interior. After m sign bisections, choose the midpoint u_m of the retained zero bracket, terminating early at any exact zero. Then

$$0 \leq \mathrm{AVaR}_\alpha(f) - R_{u_m}(f) \leq \frac{KRt}{2}\, 4^{-m}. \tag{32}$$

The right side is replaced by zero after an endpoint or exact-zero exit.

For an integer $n \geq 1$, partition each of the two selected tails into equal cells within that tail, with every cell length at most $2Rt/n$. At most $n+1$ nonempty cells are required. Midpoint values give a lower bound $\underline{R}$ for $R_{u_m}(f)$; endpoint secants give an upper bound $\overline{R}$. The resulting interval

$$\left[\underline{R},\, \overline{R} + \frac{KRt}{2} 4^{-m}\right] \tag{33}$$

contains the original AVaR and has width at most

$$KRt\left(n^{-2} + \tfrac{1}{2}4^{-m}\right). \tag{34}$$

With $m = \lceil \log_2 n \rceil$, the width is at most $3KRt/(2n^2)$ and at most $2n + 2\lceil \log_2 n \rceil + 8$ distinct point values are acquired, including tail localization.

Conversely, every deterministic adaptive interval algorithm using at most N exact point values and sound for all consistent convex K-Lipschitz functions has an input on which its reported width is at least

$$\frac{KRt}{6(N+1)^2}. \tag{35}$$

Thus, for $N \geq 32$, the optimal worst-case width is of order $KR(1-\alpha)N^{-2}$ with constants independent of $\alpha \in [0,1)$, under this exact point-information convention. At $\alpha = 0$ the construction uses ordinary midpoint–secant integration on the full interval and no localization; the same order and lower bound hold.

The lower bound is the retained indistinguishable-tail argument, with its complete proof restated in the supplement. The addition is an upper algorithm that locates the adverse interval and pays for that localization, so that the same linear tail-length factor appears on both sides. The AVaR variational principle and

midpoint–secant inequalities are classical (Rockafellar and Uryasev, 2002, Novak, 2015). This result does not make a point-value oracle equivalent to formula access. As α approaches one, representing the shrinking intervals, resolving nearby function values, and enforcing outward arithmetic may require additional bits and numerical work. Those costs are outside an exact-real point-query count and are charged in the implementation.

9.2. *An outward implementation for the original investment economy*

Fix the original investment state x, a feasible realized proposal a, the same α, and the unchanged implemented terminal controller. Let

$$f_a(z) = V_1(F(x,a,z)), \qquad Q(a) = S(x) + h(a) + \beta\,\mathrm{AVaR}_\alpha(f_a).$$

Here Q uses the original optimal last-date continuation. Its relation to the actual complete-policy cost $J(a)$ is

$$Q(a) \le J(a) \le Q(a) + \beta\varepsilon_1, \qquad \varepsilon_1 = 2^{-20}. \tag{36}$$

This allowance is inherited from the actually implemented controller; it is not inferred from the newly chosen quadrature nodes.

The primitive curvature in Proposition 4 implies that V_1 is m_x-strongly convex, with $m_x = (4 - 5\beta/4)/d$. Since the original innovation direction has squared Euclidean length d, f_a is m_z-strongly convex, where

$$m_z = 4 - 5\beta/4 = 181/64. \tag{37}$$

Its Lipschitz constant in z is at most 15. Consequently the integrated central loss H_a is $m_z L$-strongly convex. These are original-primitive constants, not fitted shape estimates.

Let any feasible $u \in I$ be supplied, with a verified interval $[d_-, d_+]$ containing $f_a(u+L) - f_a(u)$. Define

$$\kappa(u) = \frac{1}{2Rt} \max_{d\in\{d_-,d_+\}} \left[-\min_{v\in I}\{d(v-u) + \tfrac{1}{2}m_z L(v-u)^2\} \right]. \tag{38}$$

The inner minimum is attained at $v = \mathrm{proj}_I(u - d/(m_z L))$; an outward lower bound on that minimum is sufficient. Thus inaccurate endpoint comparisons enlarge a retained defect instead of granting false tail optimality.

Partition the selected tails into cells $B_i = [l_i, r_i]$ with midpoint z_i and positive weight $w_i = (r_i - l_i)/(2Rt)$. Their weights sum to one. At each successor $y_i = F(x, a, z_i)$, obtain a valid global affine minorant

$$V_1(y) \geq \ell_i + p_i^{\mathsf{T}}(y - y_i).$$

The vector p_i may itself be enclosed, provided the uncertainty is retained in the final slope. Obtain upper values u_i^- and u_i^+ for $V_1(F(x, a, l_i))$ and $V_1(F(x, a, r_i))$. Put

$$r_- = \sum_i w_i \ell_i, \qquad r_+ = \sum_i w_i (u_i^- + u_i^+)/2, \tag{39}$$

$$W = S(x) + h(a) + \beta r_-, \qquad s = h'(a) + \beta \sum_i w_i \, b^{\mathsf{T}} p_i, \tag{40}$$

where b is the original investment action-exposure vector. All sums and products must be outward enclosures of these exact weighted objects.

THEOREM 10—Original-policy qualification from a feasible active tail: *Suppose the displayed supports, value intervals, derivative interval, and scalar minimizations are valid outward bounds. Then*

$$S(x) + h(a) + \beta r_- \leq J(a) \leq S(x) + h(a) + \beta\{r_+ + \kappa(u) + \varepsilon_1\}, \tag{41}$$

$$V_0(x) \geq \inf_{0 \leq v \leq A(x)} \{W + s(v - a) + (v - a)^2\}. \tag{42}$$

If $s \in [s_-, s_+]$ is uncertain, taking the smaller of the two minima at s_- and s_+ gives a valid lower bound in (42). Subtracting this lower bound from the policy upper endpoint therefore bounds complete-policy regret relative to the original optimal policy. The original loss target $1/256$ *and policy-width target* $1/8192$ *retain their original meanings.*

The conclusion applies to boundary proposals, a one-sided tail, a nonoptimal tail cut, a rounded action, and an imperfect stored continuation used only to generate the proposal. It requires neither convexity of the implemented policy value nor differentiation through the

rounded terminal actor. With the same proposal and the same future policy, a different valid certificate changes information about cost, not the cost itself.

A cut obtained from the stored critic can be a useful proposal, but the tail defect in (38) must concern the original V_1. The same fixed feasible tail density supplies the all-action minorant; it is never differentiated as though it were an exact optimizer. The new service and the uniform-support conventional service both pay for tail proposals, original terminal evaluations, refused certificates, refinement, and fallback. Operation savings and actual returned-policy contrasts are reported separately.

9.3. *Paid precision and recursive use*

For any finite list of attempted resolutions, accepting at the first resolution whose outward intervals meet both original targets preserves qualification. Every unsuccessful earlier attempt remains part of the service cost. A geometric ladder alone gives no promise that a rejected action ever qualifies: its limiting supporting quadratic may remain too loose, or the action itself may be too costly. The unchanged fallback therefore remains part of the delivered policy.

For a declared fixed graph with exact-value quadrature allowance A/n^2, tail-localization allowance $B4^{-m}$, $A, B > 0$, positive costs $c_q n + c_b m$, and usable tolerance $\epsilon > 0$ after all other errors have been charged, an interior continuous optimum satisfies

$$e_b = \frac{2c_b}{c_q\sqrt{A}\log 4}(\epsilon - e_b)^{3/2}, \qquad n = \sqrt{A/(\epsilon - e_b)}, \qquad m = \frac{\log(B/e_b)}{\log 4}. \tag{43}$$

The equation has one root in $(0, \epsilon)$. Domain boundaries such as $n \geq 1$ or $m \geq 0$ are checked separately; upward integer rounding preserves the allowance and adds less than $c_q + c_b$. This is an allocation for the stated work model, not an empirical compute price or a globally optimal discovery policy. Its proof is in the supplement.

At earlier dates the same witness construction can be used whenever the original continuation along the innovation has the required convexity and supplies global supports. Its local original-action gaps enter the retained dated policy account; constant certified continuation allowances propagate by discounting and cash

invariance. Convexity is not asserted for an arbitrary learned network, an arbitrary nonlinear transition, or an unexecuted longer-horizon controller.

9.4. *A bounded preliminary information budget*

Resolving fewer innovation locations need not reduce complete decision work. Tail localization, several refused resolutions, and eventual repair can outweigh the saved quadrature. We therefore introduce a second service whose preliminary work is bounded before the original complete solver is invoked. Phase B freezes this algorithm after the retained Phase A outcomes; its acquisition, conventional controls, reference intervals, and timing processes constitute a separate experiment.

Fix the original state, a feasible realized actor proposal, the original risk level $\alpha \in \{1/2, 3/4\}$, and the implemented terminal policy. Two floating endpoint comparisons propose a boundary cut when possible. Only the remaining rows receive six floating bisection steps. These signs have no certification authority: the resulting feasible cut is rounded to the declared dyadic grid, and two outward original-continuation values construct the defect κ in (38). This defect is retained at every resolution. The service omits exactly zero-length tails, attempts two and then four equal cells per positive-length tail, and accepts only a certificate meeting both original targets. Otherwise it invokes the frozen 128-bin complete service from the same already acquired proposal, including its original repair, refinement, and rational fallback.

PROPOSITION 11—Original-policy qualification with bounded preliminary work: *Under the outward-oracle and original-policy assumptions of Theorem 10, the stated service returns only certificates valid for the original investment objective. Its preliminary work uses at most*

$$N_{\text{outward}} \le 30, \qquad N_{\text{floating}} \le 16 \tag{44}$$

terminal point-value payloads per proposed action before acceptance or entry to the frozen complete service. An outward payload includes the original value enclosure and the valid supporting information required by the action certificate. Calls repeated at the same location are counted again.

More precisely, if the selected density has $p \in \{1,2\}$ nonempty tails, the outward bound is $2+14p$; an acceptance at the first resolution uses at most $2+5p$. A floating endpoint exit uses only four floating values. The two outward defect values are acquired once and shared across the preliminary resolutions. If the underlying oracle calls and frozen complete service terminate, adding this finite prelude preserves that termination and the original loss and width targets.

These are bounds on added terminal information, not on elapsed time or on the complete internal work of a terminal solve. Actor acquisition, the payloads' numerical construction, rejected attempts, density and certificate algebra, and the complete invoked fallback remain paid work. Early acceptance can avoid a conventional repair and can therefore change the returned action. Its original economic cost must be compared using its own reference interval; no zero-cost or improvement conclusion follows from the point-count bound.

The masked-tail ablation applies the same exact zero-mass omission and floating proposal rule while retaining all six resolutions $2,4,8,16,32,64$. It separates the effect of removing zero-mass work from the additional decision to cap preliminary effort. Every representation, including the conventional hinge encoding of ReLU, receives the same service and information access.

9.5. *From the payload budget to original decision work*

A bound on purchased point values does not by itself bound elapsed decision work. The original compiler has a fixed cost for array construction, outward operations and its certified terminal controller. Calling it separately for each nonempty tail component can therefore lose time even when it acquires fewer original values. We implement Proposition 11's two-resolution prelude by packing its positive-weight midpoint and endpoint requests into one original compiler call per attempted resolution.

Each packed entry carries its original state, action, innovation and request type. The inverse indices return its lower value, upper value and supporting slope to the original selected-tail summand. A singleton innovation axis prevents a cross-product between different entries. Repeated locations remain separate paid requests;

zero-weight positions do not become observations. The separately cached cut-defect calculation and complete fallback retain their original costs. Thus the 30/16 preliminary information bounds and the all-action validity proof are unchanged. Supplement 16.3 proves this reindexing account, including the possibility that batch-level fallback returns different valid outward intervals. The resulting elapsed work is an empirical question measured in the separately frozen phase C.

10. POLICY IMPROVEMENT, STOPPING AND THE LEARNING GRAPH

The certificate separates three choices: which continuation to acquire, which feasible action to propose, and whether to spend more work on that action. The first two determine a candidate complete policy. A tighter enclosure of that same policy changes the information available for the third choice. Actual improvement requires a comparison of original policy values.

Completed policies and foregone gain. Let a completed feasible incumbent have a verified cost bracket $[L_B, U_B]$ and let O_B be a valid lower bound on the original optimum. A completed trial with upper cost $U_T < L_B$ delivers a strict original-policy improvement, provided it also meets the unchanged qualification targets. The improvement is at least $L_B - U_T$. The remaining possible gain from any further feasible policy is at most

$$G_B = \max\{0, U_B - O_B\}.$$

Thus stopping when $G_B \le \eta$ has an explicit original-objective opportunity cost. For a monetary value v per unit of original loss and a lower bound c on an additional computation's cost, $c \ge vG_B$ rules out a strictly positive net gain from that computation. The converse inequality does not guarantee that a useful improvement will be found. The complete original theorems, interface sharpness and recursive risk-distorted composition are retained in Development Sections K and L, with full proofs in the supplement. Their R110 experiments retain their own actual policy gains and positive foregone gains; they are not new R119 observations.

An action comparison can share a risk witness. For bounded original continuations X, Y and one common feasible dual law q, write $k_X(q) = \rho(X) - \{\mathbb{E}_q X - \psi(q)\} \ge 0$.

Then

$$\rho(Y) - \rho(X) = \mathbb{E}_q(Y - X) + k_Y(q) - k_X(q).$$

Supplement Theorem 5 proves a computable action contrast using the two original-risk defects, with an atom-safe quantile-band remainder. Its implemented-future allowance is charged once on each side of the contrast. Proposition 6 proves that a continuation contrast of oscillation $w > 0$ only needs dual laws whose reference optimality slack is at most w, without assuming the supremum is attained. This preserves the signed recursive policy difference instead of discarding the reference slack. The integral of a difference of convex continuations still needs a valid separate enclosure. This mathematical interface advances the independently frozen R111–R112 question; no current numerical result is attributed to a paired algorithm that has not been implemented.

The actor and critic are a specified computational graph. For a smooth critic u_θ and a differentiable feasible actor a_ω, the Bellman residual includes the derivative through the actor and through every parameter-dependent continuation and transition. Differentiating the half-squared residual gives $\mathbb{E}[r_{\theta,\omega} \nabla r_{\theta,\omega}]$ under the integrability and differentiation hypotheses in Supplement Section 13. Holding a target fixed defines a different update and requires its own convergence argument. Discrete original-law qualification treats a realized rounded action as a feasible decision and does not differentiate its numerical rounding operation.

For the Brownian Markov recursive-utility interface, the critic graph is $z_u = \sigma^\top \nabla u$. The driver, admissible controls, terminal condition and stopping boundary belong to that same recursive criterion. Supplement Proposition 4 proves the stopped-domain residual account, including its stability factor and boundary discrepancy. A general ReLU critic is not a classical $C^{1,2}$ Itô test function merely because its ordinary Hessian vanishes away from kinks. The discrete NBO construction remains valid under its own hypotheses. Fully coupled NDU/FBSDE systems and other utility objectives require their own well-posedness and verification premises; they do not inherit the investment experiment's numerical ranking by a change of terminology.

11. ORIGINAL INVESTMENT DECISIONS: INFORMATION AND COMPLETE COST

11.1. *The economic comparison and its information budget*

We evaluate whether resolving the active innovation tails reduces the cost of a certified original Bellman decision. The economic model remains (13)–(14): two investment dates, a nonlinear transition, the original state-dependent investment capacity, discount $15/16$, a continuous uniform innovation on $[-1/32, 1/32]$, and nested upper-cost AVaR at $\alpha = 1/2$ and $3/4$. The actual last-date controller, its observation rule, and its 2^{-20} uniform continuation allowance are unchanged. At the first date every completed service must certify regret at most $1/256$ and an original-policy cost interval of width at most $1/8192$. Theorem 10 establishes why these criteria concern the original economic objective, including the implemented future decisions. Numerical integration encloses the original continuous law throughout.

The intervention holds each acquired continuation representation and its realized first-date proposal fixed while changing the information used for qualification. The eight arms are the online semi-analytic Bellman control, quadratic, ReLU, its equivalent conventional hinge, coordinate spline, degree-two and degree-three polynomials, and grid interpolation. The rank-six quadratic, ReLU and spline constructions retain their common 32-feature dictionaries, three nonlinear selections, nonnegative nonlinear coefficients, tie rule and 30-bit rounding. The broader polynomial and grid baselines retain their additional representational flexibility and reported memory requirements. The hinge receives exactly the acquired ReLU coefficients and its full standalone acquisition charge. It provides a conventional implementation of the same information and feasible policies.

In each phase we newly acquire the 256 original-model terminal-value labels on the fixed 16×16 grid for each risk level. Every competing fit receives the same label bytes. Acquisition includes all candidates, exclusions, independent feature-matrix and active-face QR audits, serialization and reloading. The primary catalogue is all 144 states of the fixed 12×12 grid. These are deterministic numerical catalogues; no population confidence interval or unseen-sample claim attaches to them. The experiment concerns the acquired last-continuation slice of NBO in this original

economy. It does not supply a new trained long-horizon or empirically calibrated application.

Phase A froze its executable protocol at `1f33c266` before its new acquisition and execution. It compared the complete 128-bin compiled service (C), the retained geometric whole-support schedule (G), and the new active-tail schedule (T), producing 6,912 primary decisions and 10,752 timing decisions. Each service pays for its proposal, every rejected certificate, refinement and conventional repair. All 128-bin fallbacks and eventual exact fallbacks remain available. The seven fresh timing processes each use the first 32 catalogue states, with service order rotated by the declared rule. All clocks, including the disclosed document-formatting overlap during phase A, remain archived without replacement.

Phase A exposed a specific computational problem. Active-tail qualification reduced original terminal-risk queries from 654,836 to 293,844 over the full primary catalogue, but the geometric service required only 223,767. It beat the compiled service in 12 of 16 median-clock comparisons and the geometric service in only one; the latter ratio was 0.9947, with only three of seven paired clocks lower. ReLU was slower than both controls at both risk levels in every recorded repeat. The full 48-row frontier, including these losses, remains in the development companion.

11.2. *A bounded preliminary certificate and complete services*

Phase B was designed after observing phase A and prospectively committed at `357b2c56` before its scientific execution. It is disclosed development on an already observed state catalogue. It freshly repeated acquisition, all three controls, reference work and every timing process; phase-A observations were not used as phase-B controls. Two additional services isolate the implementation and precision mechanisms. The masked service (M) omits zero-mass tail queries and floating interior searches after a boundary exit, retaining all six resolutions $2, 4, 8, 16, 32, 64$. The budgeted service (B) uses the same masking and original-policy certificate, but permits only resolutions 2 and 4 before the unchanged 128-bin gate and complete conventional fallback. It bounds preliminary outward terminal-value payloads by 30 per decision, or 16 for a boundary tail, and floating proposal values by 16. The

TABLE I

PHASE B: COMPLETE 32-STATE SERVICE TIMES, ALL ARMS AND SERVICES

α	Representation	C	G	T	M	B
1/2	Online	0.1015	0.0778	0.0895	0.1107	0.1096
1/2	Quadratic	0.0588	0.0336	0.0422	0.0650	0.0646
1/2	ReLU	0.0896	0.0965	0.1187	0.1361	0.1071
1/2	Hinge	0.0819	0.0865	0.1097	0.1256	0.0957
1/2	Spline	0.0582	0.0318	0.0417	0.0611	0.0604
1/2	Polynomial 2	0.2227	0.2022	0.2170	0.2353	0.2298
1/2	Polynomial 3	0.2864	0.2509	0.2665	0.2901	0.2858
1/2	Grid	0.2343	0.1983	0.2091	0.2342	0.2269
3/4	Online	0.1030	0.0761	0.0761	0.0881	0.0873
3/4	Quadratic	0.0584	0.0318	0.0324	0.0431	0.0429
3/4	ReLU	0.0904	0.0969	0.1182	0.1189	0.0934
3/4	Hinge	0.0801	0.0848	0.1076	0.1119	0.0835
3/4	Spline	0.0585	0.0317	0.0324	0.0419	0.0430
3/4	Polynomial 2	0.2190	0.1853	0.1870	0.1984	0.2182
3/4	Polynomial 3	0.2655	0.2398	0.2420	0.2515	0.2485
3/4	Grid	0.2262	0.1936	0.1981	0.2071	0.2127

Medians in seconds from all seven fresh sequential processes. C: 128-bin compiled; G: geometric whole-support; T: original active tail; M: positive-tail masking with the full ladder; B: masked two-level budget. Every service pays its proposal, refusals, repairs and certificates. Acquisition, actual loading and enclosing process costs are reported separately. Observed ranges and every clock are retained in the development companion.

bounds include the two outward endpoint payloads that certify the selected tail's true AVaR defect. Floating signs never grant certificate authority.

All 11,520 primary and 17,920 timing decisions completed. Independent record arithmetic verified 1,031,933 claims over 45,557 attempted certificates. A separate read-only reconstruction checked all 640 raw file hashes and 295,161 additional qualification, policy-identity and payload-account claims. The two-level budget attained but never exceeded 30 outward values, and attained but never exceeded 16 at a boundary, over all 5,888 primary and timing rows of that service. It used 28,801

preliminary outward payloads in the primary catalogue, compared with 94,925 for the full masked ladder. The 244 primary proposals exhausting that ladder were the same 244 requiring the complete fallback under the two-level rule. All were retained and paid.

Table I reports every arm, risk and service. Masking alone reduced risk queries from 293,844 to 220,341, yet was slower than the original active-tail implementation in all 16 median comparisons. The budgeted rule reduced them further to 154,217, but beat geometric whole-support qualification in only two of 16 medians. These were ReLU and the equivalent hinge at $\alpha = 3/4$, with ratios 0.9640 and 0.9843; their seven paired-clock win counts were seven and five. ReLU at $\alpha = 1/2$ instead had ratio 1.1093 and lost in all seven repeats. Thus lower information demand and fewer preliminary resolutions do not establish a generally cheaper implementation. Mask grouping, interval assembly, recorded traces, proposal inference and complete fallback remain actual service work.

The 144-state workload gives a related but different comparison. Budgeted qualification was faster than geometric qualification in seven of 16 cells and faster than compiled qualification in all 16. Summing the observed primary service times across all arms and risks gives 9.2482, 7.2390, 8.0358, 8.6385, and 7.5186 seconds for C, G, T, M, and B. These totals retain all cells and are not a selectively composed service. The small-batch and full-catalogue rankings need not agree: the first 32 states contain a different incidence of costly initial proposals, and vectorized grouping has a workload-dependent cost. We do not pool their clocks or assign an unobserved request distribution.

11.3. *Original policy costs, action error and adoption*

The reference calculation partitions the full original innovation support into 1,024 bins independently of the qualification mesh and evaluates every distinct returned state-action policy. It shares the verified terminal compiler with production; it is not a fully independent implementation. The separately implemented exact-rational primitive suite reconstructs 24 original reference cases and 72 certificates, and has that finite scope. It pays for 983 distinct policy queries and 9.6771 seconds in phase

B. The best valid all-action lower bound and feasible policy upper bound at each state enclose the original optimum. They give original-policy regret intervals and action-distance bounds using original action curvature two. The largest reported regret upper bound is 0.00285149, below $1/256$; the largest action-distance upper bound is 0.053400. These are verified bounds, not exact measured action errors.

Every one of the ten within-arm service comparisons has exactly the same returned complete policy at all 2,304 state–risk–arm combinations. Its original economic cost contrast is therefore exactly zero. The 1,220 recorded primary action repairs are 244 distinct initial proposals paid and repaired separately under five services. No exact rational fallback was required. More informative certificates change the information required to serve these policies; this experiment supplies no new policy improvement event. The distinct original-policy improvements of the retained R110 safeguard and gain-budget campaigns remain reported with their own sources, catalogues and clocks in Section R.

The conventional frontier is economically material. ReLU's mean original cost exceeds quadratic's by an interval of $[1.79075, 1.87557] \times 10^{-5}$ at $1/2$ and $[1.80271, 1.88385] \times 10^{-5}$ at $3/4$. ReLU is individually cheaper at 43 and 39 states, but the whole-catalogue mean differences are strictly adverse. Against online control, it is identical at 108 states and strictly costlier at 36 under either risk level. Its mean cost exceeds grid interpolation's and is below spline's and degree-three polynomial's; every pairwise interval, including unresolved statewise comparisons, is retained. The exact ReLU–hinge identity holds for 80 primary/timing record pairs containing 3,680 paired decisions and all their attempted certificates. Their differing elapsed times cannot be attributed to an exclusive neural representation.

Table II includes each arm's actual standalone acquisition and complete 144-state cost. Quadratic with geometric qualification costs 0.1648 and 0.1620 seconds, versus 0.3755 and 0.3642 for the budgeted ReLU service. It also has the lower certified catalogue-mean economic cost. For this entire 144-state workload the ReLU acquisition saving is insufficient to offset service work. This observation leaves neural adoption unestablished on the measured frontier. The inherited least-squares acquisition criterion also remains economically imperfect: the degree-three fit has

TABLE II

PHASE B: PAID ACQUISITION, COMPLETE 144-STATE COST AND ECONOMIC BOUNDS

α	Representation	A	F_G	F_B	$10^6\overline{L}$	$10^3\overline{d}_{\max}$
1/2	Online	0.0001	0.2960	0.3496	0.899	0.957
1/2	Quadratic	0.0634	0.1648	0.1874	153.657	31.496
1/2	ReLU	0.0599	0.3927	0.3755	171.989	51.389
1/2	Hinge	0.0602	0.3341	0.3182	171.989	51.389
1/2	Spline	0.0536	0.3274	0.3395	264.595	53.400
1/2	Polynomial 2	0.0105	0.6376	0.8062	153.657	31.495
1/2	Polynomial 3	0.0107	0.7882	0.9014	215.969	29.648
1/2	Grid	0.0101	0.8293	0.7160	11.069	11.106
3/4	Online	0.0002	0.2823	0.3395	0.900	1.022
3/4	Quadratic	0.0698	0.1620	0.1796	163.145	31.487
3/4	ReLU	0.0638	0.4118	0.3642	181.578	49.019
3/4	Hinge	0.0639	0.3765	0.3436	181.578	49.019
3/4	Spline	0.0603	0.3851	0.3464	289.860	52.111
3/4	Polynomial 2	0.0154	0.7420	0.8229	163.147	31.487
3/4	Polynomial 3	0.0155	0.8937	0.9413	209.222	30.155
3/4	Grid	0.0150	0.7887	0.7603	35.811	18.471

A includes original labels, every fit and candidate audit, and serialization; hinge includes the full ReLU acquisition. $F_s = A +$ actual load time $+$ actual 144-state service time, in seconds. These are recorded component sums, not new clocks. $\overline{L}$ is an outward upper bound for catalogue-mean original-policy regret; $\overline{d}_{\max}$ bounds the largest distance to the original optimal action. All five services have the same returned policies, so their economic bounds coincide. Displayed bound endpoints are rounded outward.

lower terminal fitting loss than quadratic, while its original catalogue-mean regret upper bound is higher. New certificates do not turn this into action-aligned training.

For a deployment repeating this same 144-state catalogue, let A_j include acquisition and one model load, C_j the complete per-catalogue service cost, and ΔJ the mean original-policy cost difference from a specified comparator. With H catalogue repetitions and a specified value $\lambda \geq 0$ per economic-cost unit in the same accounting units, a sufficient adoption condition is

$$A_j - A_c + H(C_j - C_c) + 144H\lambda\,\overline{\Delta J} < 0.$$

It uses an upper economic contrast bound, not a difference of regret ceilings. No H, compute price, monetary calibration or request population is supplied by the experiment. At fixed identical policies the economic term is zero. Measured information savings therefore justify adoption only when complete implementation cost, acquisition and the intended workload satisfy the corresponding condition.

11.4. *Information theory and the scope of scaling evidence*

The separate exact point-information catalogue tests Theorem 9 on ten predeclared convex piecewise-linear/quadratic functions, eight risk levels from zero through 255/256, and four resolutions. All 320 cells and 2,901 rational checks are retained. The active-tail interval is narrower in 242 cells, whole-support quadrature is narrower in 27, and 51 tie. Both methods obey the same declared query cap; actual distinct queries are reported separately. At $\alpha = 0$, whole-support wins 20 of 40 cells and active integration wins eight. Nearer the upper risk limit active localization uses the shorter adverse support, consistently with the theorem's $1 - \alpha$ factor. Exact independent integrals and cut reconstruction are evaluation evidence; formula access is not provided to either point-query algorithm.

Phase A also compares the same original online proposal with a compiled 128-bin certificate and an eight-cell-per-tail certificate at dimensions $2, 4, 8$. All 192 paired contexts qualify. The active/compiled certificate time ratios are approximately $0.515, 0.475$, $0.423, 0.424$, and $0.391, 0.317$ across the two risks. Charging their common proposal moves the ratios to $0.829, 0.793$, $0.757, 0.763$, and $0.690, 0.659$. These are original-model primitive scaling observations. They do not establish a complete learned high-dimensional service, a multivariate risk integration theorem, or a longer-horizon convergence rate. The mathematical information improvement, its action-valid certificate, and these implementation limitations together specify the next economic computational problem; all historical adverse and incomplete episodes remain attributable in the development companion.

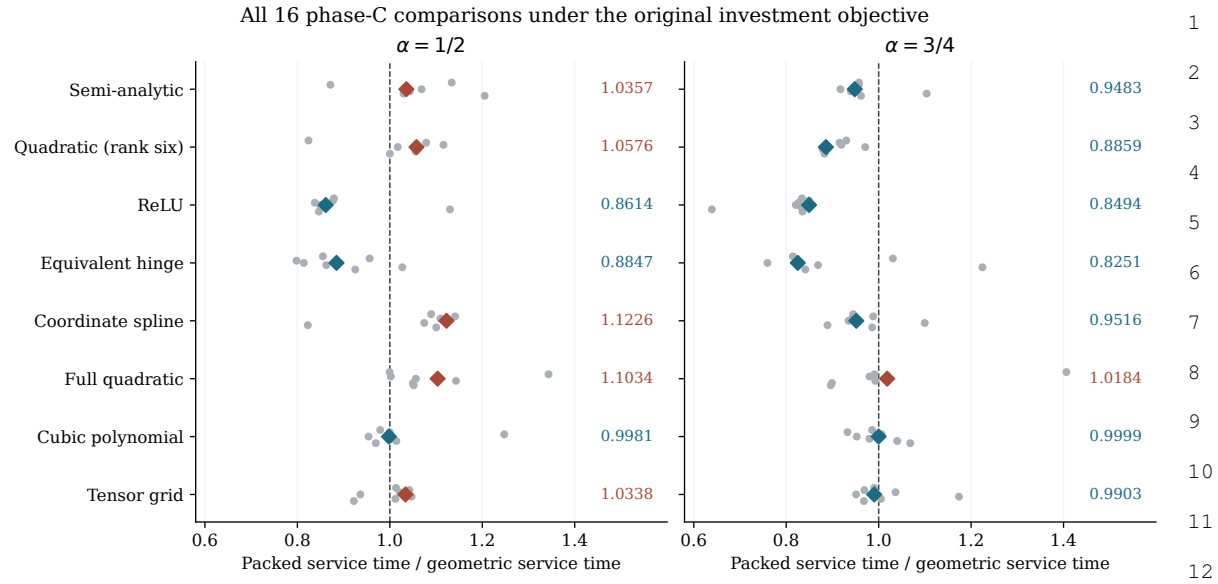


FIGURE 1.—All contemporaneous phase-C packed/geometric service comparisons. Diamonds show the ratio of the two seven-process median times; grey points show every paired process ratio. Values below one favor packing. These are 32-decision service clocks with actual loading and broader process costs reported separately, not confidence intervals.

11.5. *Packed original-law evaluation: a separately frozen implementation*

The negative masking result motivates a further implementation experiment, phase C, prospectively committed at `095936f1` after the outcomes of phases A and B were known. It repeats all five controls, acquisition, reference calculations and seven timing processes afresh. The sixth service, fused tail (F), purchases exactly the positive-mass midpoint and endpoint payloads required by the two-level budget, in one packed call to the unchanged original terminal compiler at each resolution. It then scatters the returned values and slopes to their original state/action identities and performs the same outward aggregation. No duplicate requested payload is made free. The density, certified cut defect, 2^{-20} implemented-continuation allowance, original action interval, qualification targets and complete fallback remain unchanged. This is post-A/B development on the previously observed catalogue, not an unseen validation or a retrospective alteration of either earlier experiment.

All 13,824 primary and 21,504 timing decisions completed. The native exact record audit passes 1,325,935 checks over 55,043 attempted certificates; a separate

TABLE III

PHASE C: COMPLETE 32-STATE SERVICE TIMES, ALL ARMS AND SERVICES

α	Representation	C	G	T	M	B	F
1/2	Online	0.1030	0.0767	0.0852	0.1048	0.1074	0.0794
1/2	Quadratic	0.0586	0.0332	0.0412	0.0627	0.0621	0.0351
1/2	ReLU	0.0901	0.0937	0.1215	0.1354	0.1093	0.0807
1/2	Hinge	0.0823	0.0867	0.1113	0.1278	0.0998	0.0767
1/2	Spline	0.0569	0.0319	0.0425	0.0639	0.0638	0.0358
1/2	Polynomial 2	0.2146	0.1917	0.1998	0.2165	0.2194	0.2115
1/2	Polynomial 3	0.2796	0.2520	0.2580	0.2852	0.2892	0.2516
1/2	Grid	0.2265	0.1935	0.2057	0.2276	0.2266	0.2001
3/4	Online	0.0983	0.0750	0.0753	0.0867	0.0857	0.0712
3/4	Quadratic	0.0570	0.0334	0.0324	0.0428	0.0419	0.0296
3/4	ReLU	0.0884	0.0936	0.1164	0.1207	0.0954	0.0795
3/4	Hinge	0.0789	0.0872	0.1075	0.1153	0.0848	0.0720
3/4	Spline	0.0596	0.0331	0.0330	0.0433	0.0436	0.0315
3/4	Polynomial 2	0.2154	0.1917	0.1918	0.2046	0.2106	0.1952
3/4	Polynomial 3	0.2679	0.2403	0.2450	0.2541	0.2563	0.2402
3/4	Grid	0.2165	0.1950	0.2022	0.2040	0.2098	0.1931

Medians in seconds from all seven fresh sequential processes. C: 128-bin compiled; G: geometric whole-support; T: original active tail; M: positive-tail masking with the full ladder; B: masked two-level budget; F: the same budget with one packed compiler call per resolution. Every service pays its proposal, refusals, repairs and certificates. Acquisition, actual loading and enclosing process costs are reported separately. Observed ranges and every clock are retained in the development companion.

read-only report verifies all 768 raw hashes and 384,551 further exact/identity conditions. An additional 21,974 read-only checks reconstruct the same original point coordinates for the budgeted and fused services from their exact state, action, cut and resolution records. All 5,888 paired decisions use identical paid preliminary information and qualification exits. F uses 27 packed compiler calls for the primary catalogue and 182 across the seven timing processes, containing respectively 24,193 and 35,504 midpoint/endpoint payloads. The separate two-point defect calculation remains charged. The 30-value general and 16-value boundary preliminary caps

are attained and respected. Complete original risk-point queries remain 154,217, exactly the budgeted service's count.

Table III reports the full contemporaneous comparison. F is faster than B in all 16 median-clock cells, including 11 with all seven paired clocks lower. It is faster than C in all 16 medians, than T in 14, and than G in ten. The remaining six losses against G are retained. At $\alpha = 1/2$, the ReLU ratio F/G is 0.8614, with six paired wins; at $3/4$ it is 0.8494, with seven. Conventional hinges benefit similarly. Quadratic at $1/2$ instead has ratio 1.0576, while quadratic at $3/4$ has ratio 0.8859 and seven paired wins. The degree-three polynomial ratios 0.9981 and 0.9999 illustrate why a median win alone is weak evidence of a substantial advantage. All ranges and every clock remain reported; no timing confidence interval or architecture-independent speed guarantee is inferred.

Across all primary cells, F costs 6.5522 seconds, compared with 6.7726 for G and 6.8250 for B. It is faster than G in 11 of 16 complete 144-state service cells and slower in five. These totals describe the entire predeclared tournament. They do not construct a new service by selecting the fastest implementation state by state. The result identifies a practical part of the information-to-work connection: the proved small payload budget becomes more competitive when its many individual arrays are evaluated together. Packing, scattering, outward zero terms, attempted certificates and all fallback work remain inside the measured boundary.

All fifteen within-arm service contrasts still have exactly the same complete original policy at each of 2,304 state–risk–arm combinations. Their original economic contrast is exactly zero. Repacking changes some outward endpoints—600 primary and 2,030 timing rows differ from B—without changing their qualification exits or policies. Original regret and action-distance bounds retain their previous interpretation, and no exact rational fallback is needed. The 1,464 primary repairs are the same 244 distinct proposals repaired under six services.

Representation-specific adoption remains unsupported. Charging fresh standalone acquisition and actual loading, quadratic with G costs 0.1485 seconds for the full $\alpha = 1/2$ catalogue, versus 0.3407 for ReLU with F. At $3/4$, quadratic with F costs 0.1417, versus 0.3131 for ReLU with F. Quadratic also has the strictly lower catalogue-mean original economic cost at both risks. The ReLU–hinge identity holds

in all 96 paired files and 4,416 paired decisions, including attempted certificates. The conditional adoption inequality in Section 11 therefore continues to require the full resource and economic comparison, not an architecture label. Phase C adds a shared original-decision computational improvement while preserving the negative A/B implementation findings, R97's adverse acquisition result, and the still-open action-aligned learning and larger-application obligations.

11.6. *Independent value enclosures for every recorded original policy*

The 1,024-bin reference uses a separate whole-support risk partition with the shared verified terminal compiler. To check original policy values through a separate implementation, validation D was frozen at `cafd38ae` after all A–C outcomes were known. It fixes 64 whole-support cells and up to 32 rational terminal bisections on every one of the 983 realized state/action/risk keys represented by C's 13,824 primary decisions. The exact binary state and action identities are retained. This is validation of the known catalogue.

The standard-library implementation reconstructs the original nonlinear economic primitives, the terminal shock polynomial and its continuous AVaR integral, and a feasible strong-convexity bracket for V_1^*. Conditional Jensen bounds for $(V_1^* \circ F - \eta)_+$ and the repeated-endpoint secant mixture give independent whole-support bounds for the original Q. Proposition 7 in the supplement proves these bounds. Adding the unchanged $\beta 2^{-20}$ future-controller allowance once gives the complete-policy J enclosure. The production compiler is not imported. This does not directly evaluate the deployed floating terminal controller or independently determine the global date-zero regret optimum.

All 983 keys and 126,807 exact terminal contexts completed, with 652,774 recorded checks and no failed key. The independent intervals overlap all 13,824 corresponding service intervals and all 983 shared-compiler reference intervals. Overlap establishes compatibility; it does not certify each fine reference endpoint merely from intersection. All exact traces, checks and containment directions are retained.

All 96 arm/service/risk means and all 56 arm contrasts are independently reaggregated. There are 32 strictly negative, 20 strictly positive, 2 exact-zero and 2 unre-

TABLE IV

INDEPENDENT FRACTION ORIGINAL-POLICY MEAN-COST ENCLOSURES ON THE FIXED CATALOGUE.

α	Representation	Lower mean cost	Upper mean cost
1/2	Semi-analytic	0.611452244	0.611454152
1/2	Quadratic (rank six)	0.611604999	0.611606910
1/2	ReLU	0.611623342	0.611625227
1/2	Equivalent hinge	0.611623342	0.611625227
1/2	Coordinate spline	0.611715940	0.611717849
1/2	Full quadratic	0.611604999	0.611606910
1/2	Cubic polynomial	0.611667314	0.611669224
1/2	Tensor grid	0.611462422	0.611464314
3/4	Semi-analytic	0.621104706	0.621106644
3/4	Quadratic (rank six)	0.621266923	0.621268911
3/4	ReLU	0.621285373	0.621287354
3/4	Equivalent hinge	0.621285373	0.621287354
3/4	Coordinate spline	0.621393664	0.621395613
3/4	Full quadratic	0.621266925	0.621268914
3/4	Cubic polynomial	0.621313002	0.621315001
3/4	Tensor grid	0.621139623	0.621141543

Each endpoint is an outward decimal display of an exact rational bound. Every mean weights the same 144 realized original states equally. All six services return exactly the same complete policy within an arm; all 96 arm/service/risk means are retained in the development companion. The future-controller allowance is $\beta 2^{-20}$. These intervals are numerical enclosures, not sampling confidence intervals.

solved contrast intervals in the fixed left-minus-right orientation. The ReLU-minus-quadratic original mean-cost intervals are $[17.4360, 19.2238] \times 10^{-6}$ at $\alpha = 1/2$ and $[17.5331, 19.3595] \times 10^{-6}$ at $\alpha = 3/4$. The full contrast table retains every sign and exact tie. These independent intervals confirm the rank-six quadratic arm has strictly lower mean original policy cost than ReLU at both risk levels. ReLU and its exactly equivalent conventional hinge representation retain identical realized policies and exact-zero mean cost contrasts. All 240 arm/risk/service-pair cells have identical

complete policies, preserving the exact zero economic effect of changing only the certification service. No favorable economic sign was a validation pass condition.

D requires 1697.332 seconds of additional actual author-side validation time (1693.604 CPU seconds), including its separately recorded binding, rational solves, primitive traces and aggregation. This cost is reported separately from the unchanged A–C acquisition and service clocks. D strengthens independent original-model value evidence; it creates neither an unseen population result nor a neural-specific adoption claim.

12. CONCLUSION

Neural Bellman Operators link recursively acquired continuations to feasible original-objective decisions. The complete account includes the actor's own future policy, action-dependent continuation error, original-law numerical witnesses and the cost of obtaining them. The original construction and the completed-policy safeguards establish precise decision guarantees without assuming that an optimizer successfully trains an arbitrary network.

The present revision identifies the adverse innovation region as a paid information decision. For scalar uniform risk and convex Lipschitz continuations, locating that region before integration yields matching deterministic bounds with linear tail-length dependence. The original investment primitives turn an approximate location into an outward all-action certificate, and a finite preliminary budget limits work spent before the complete original solver is invoked. These contributions concern the information and certification components of NBOs and are available to all same-information representations.

The experiments distinguish information savings, service time and economic loss. They retain every fixed-catalogue decision, every timing repetition, all rejected certificates, independently paid references and the strong conventional frontier. Equivalent ReLU and hinge actors receive equal information and full acquisition charges. An unchanged complete policy has unchanged original economic value. Computational adoption therefore requires both a competitive complete service and a justified economic workload; a smaller fitting residual or a smaller query count alone does not establish it.

The research program continues toward prospective action-aware acquisition, neural-specific benefits under matched resources, larger dynamic economic problems and calibrated use. The new proofs and executed algorithms make those questions more sharply testable. The complete supplement states the mathematical assumptions and proofs, while the development companion retains the earlier applications, policy improvements, negative experiments, unresolved claims and independent source identities needed to continue that program.

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